

AE 4451 Project

Air Breathing Engine Design Project

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Section I: Executive Summary

This report goes over a goal of performing jet engine cycle analysis for the purpose of designing optimal input conditions for given flight conditions and thrust performance requirements of a commercial airliner. There are three distinct sections of this project. The first is designing two engines; one that can reach thrust requirements for ground roll flight condition, and one that can reach thrust requirements for a high altitude flight condition. The second section is designing a single engine cycle that can fulfill both flight conditions and minimize specific fuel consumption. The third section is finding the effective maximum thrust based on the second engine.

Based on initial analysis of engines, it was determined that the main focuses would be low to high-bypass turbofan engines and turbojet engines. Using various plots of thrust specific fuel consumption and checking the equations with the test case, it was found that a turbofan engine was able to complete the specific thrust requirement for the ground roll condition. The code is able to calculate temperature and pressure states through the diffuser, fan, compressor, main burner, turbine, and afterburner using standard polytropic relations. For the high altitude flight condition, similar calculations were done, with the exception of the afterburner.

The second objective identified a single engine cycle that would satisfy thrust requirements at both flight conditions using varying throttle settings. We selected an afterburning medium-bypass turbofan and it matched the specific thrust requirements of $P_{rc} = 22$, $P_{rf} = 1.95$, and $\beta = 4.5$. This engine offers a good balance between thrust capability and cruise efficiency.

The final part of this project focused on evaluating how much performance margin the optimized afterburning medium-bypass turbofan (AB-MBTF) could deliver when pushed to its maximum allowable operating temperatures. Using the Goal-2 optimized parameters as the fixed hardware configuration, a MATLAB-based throttle sweep was performed.

For the ground-roll condition, the engine specific thrust was $3.0176 \text{ kN}\cdot\text{s}/\text{kg}$, slightly above the required $2.8 \text{ kN}\cdot\text{s}/\text{kg}$. The main burner's operating temperature T_{04} was equal to 1900 K , and the temperature of the afterburner T_{06} was equal to 2600 K . Both of these temperatures are near the material temperature limit assumption. The TSFC for this condition was $2.31 \times 10^{-5} \text{ kg}/\text{N}\cdot\text{s}$, which shows a significant fuel penalty that is expected when operating with an afterburner, and is consistent with the high thermal inputs at these conditions. The results match the numerical values reported in Table IV.

A similar throttle sweep was performed at the cruise condition ($M = 0.86$, $T_a = 220 \text{ K}$, $p_a = 29 \text{ kPa}$). Applying the same strategy as the previous condition, it was revealed that the AB-MBTF could produce a specific thrust of $2.8564 \text{ kN}\cdot\text{s}/\text{kg}$, substantially above the cruise specific thrust requirement of $0.86 \text{ kN}\cdot\text{s}/\text{kg}$. At these elevated temperatures, the TSFC increased to $2.47 \times 10^{-5} \text{ kg}/\text{N}\cdot\text{s}$, indicating the trade-off between an increase in thrust and fuel efficiency, as expected. Overall, this step demonstrated that the AB-MBTF design not only met the manufacturer's requirements, but also had a substantial thrust margin at both operating conditions.

Section II: Introduction

A manufacturer requires a commercial airliner with a few propulsion systems design, each for different input scenarios. The two flight conditions are characterized by flight condition, ambient temperature, pressure, flight Mach number, and a specific thrust requirement. The properties are listed in Table I.

Table I. Flight Conditions and Thrust Requirements of New Vehicle

Vehicle	Flight Condition	T_a (K)	p_a (kPa)	M	Required Specific Thrust (kNs/kg)
Commercial Airliner	Ground Roll (Static Thrust)	285	100	0	2.8
	High Altitude Subsonic Cruise	220	29	0.86	0.86

The manufacturer ultimately desires an ideal engine cycle. To work towards this goal, three tasks were assigned. Firstly, two engine cycles were designed: one to be ideal for the ground roll (static thrust) condition, and one to be ideal for the high altitude subsonic cruise condition. The “ideal” engine for each is laid out in Table I, with the requirement being a required specific thrust based on some inputs. The cycles, while trying to meet this condition, were also designed to maintain high efficiency in the form of low specific fuel consumption. The second task was to determine one cycle that could meet the requirement of both flight conditions and also keep in mind the importance of high efficiency. The third goal was to find the maximum thrust of the engine cycle used for the second goal. The components are fixed so that the type of engine/burner cannot be changed, but fuel/air ratios and bleed ratios can be optimized to generate maximum thrust.

To complete the analysis, we developed a general engine-cycle model in MATLAB capable of representing the turbojet and turbofan configurations. The equations are summarized in Appendix A and the code that implements this full cycle simulation is shown in Appendix C. The derivations for the listed equations are listed in Appendix B. Using these equations and this program, an optimal engine was determined for each task using the progression outlined in the next section. For this project, since there are three distinct goals from the manufacturer, the code is split up into three large blocks, for easy access to each goal.

Section III: Results and Discussion

When designing the engines required by the manufacturer, the two distinct operating conditions were kept in mind: a ground-roll takeoff condition (which required a specific thrust of 2.8 kNs/kg) and a high-altitude subsonic cruise condition (which required a specific thrust of 0.86 kNs/kg). The conditions were required to have been met based on temperature, pressure, and mach number conditions. For both conditions, an engine was designed based on meeting the minimum thrust requirement, minimal parameter changes, optimization of bypass ratio, fuel-air ratio, and compressor/fan ratios. The design parameters that were available to change were compressor pressure ratio, fan pressure ratio, bypass ratio, bleed fraction, burner fuel-air ratio, and afterburner fuel-air ratio.

Ground Roll Engine Selection

For the ground roll case, in order to fulfill the high specific thrust requirement listed in Table I, a turbojet and an afterburning turbofan were analyzed. Figure 1 compares the specific thrust that can be reached with each engine, depending on the afterburner fuel-air ratio. MATLAB optimization was used to determine the best value for the fuel-air ratio, based on changing inputs. A MATLAB-based optimization sweep was used for a range of fuel-air ratios from 0 to 0.04. As the plot in Figure 1 shows, across a wide range of fuel-air ratios, the afterburning turbofan provided a much higher specific thrust at lower fuel-air ratio relative to a turbojet.

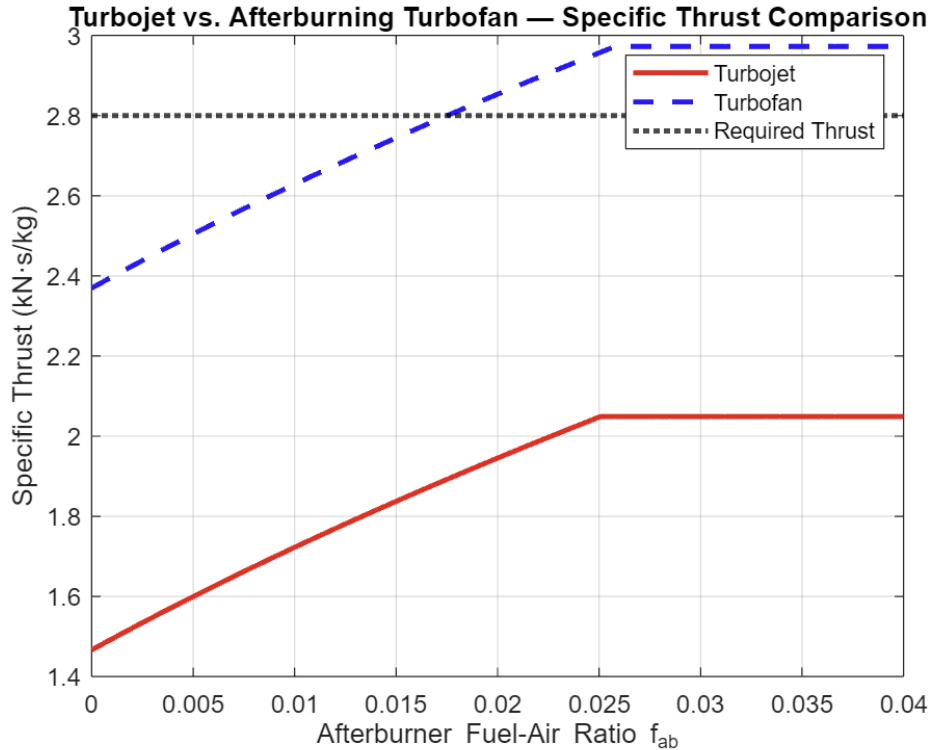


Figure 1. Turbojet vs. Turbofan for Ground Roll Specific Thrust Requirement

The figure shows that the turbofan design and the required thrust intersect at an afterburning fuel-air ratio of 0.0176, meaning that the required specific thrust of 2.8 kNs/kg can be met with an afterburning turbofan. Table II shows the input values that were input to achieve this thrust. Since the required specific thrust could be met, this engine cycle was chosen for the ground-roll condition. The afterburner provided the necessary boost during takeoff without requiring high compressor/fan pressure ratios.

High-Altitude Cruise Engine Selection

Similarly, for the high-altitude cruise condition, in order to fulfill the specific thrust requirement listed in Table I, both a turbojet and turbofan were analyzed. In the same manner as above, the team determined that for every fuel-air ratio case, the turbofan was more efficient and was able to meet the specific thrust requirement. At cruise, fuel efficiency becomes more important than the thrust, so an afterburner is not relevant due to its high fuel consumption (and in many cases, thermal drawbacks). Thus, for the high-altitude cruise condition, the afterburner was not considered.

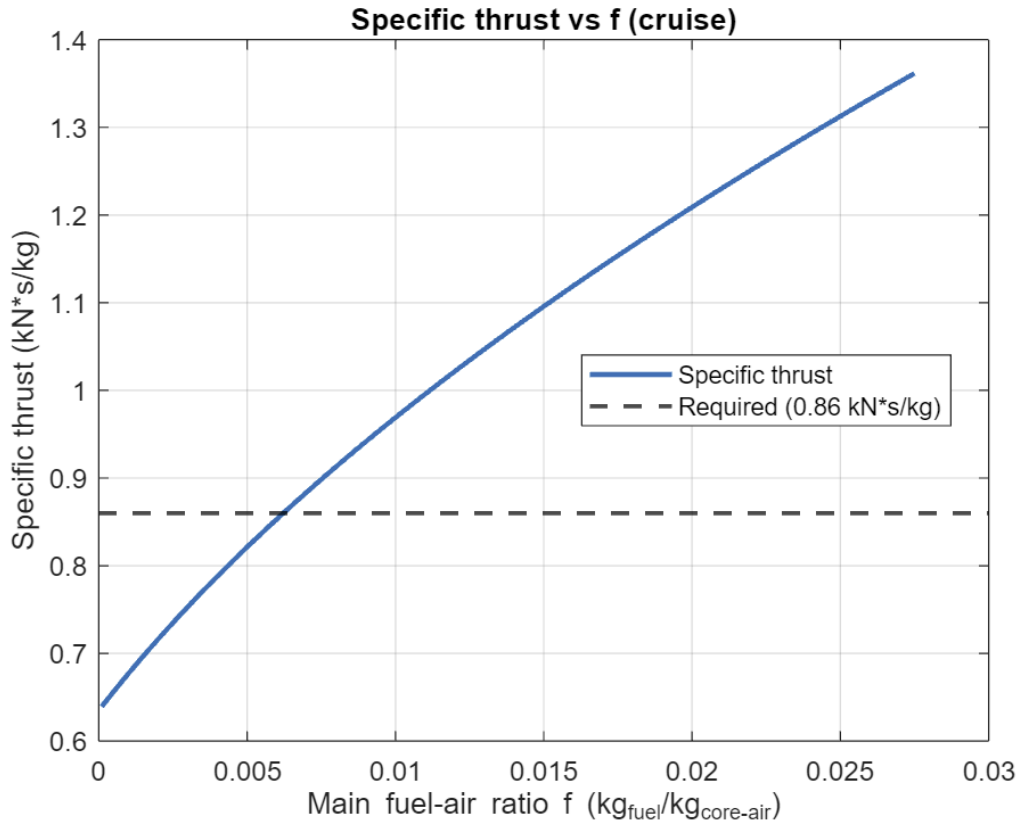


Figure 2. Specific Thrust vs. Fuel-Air Ratio for High-Altitude Subsonic Cruise Condition

A specific-thrust sweep, shown in Figure 2, was performed to optimize and determine the fuel-air ratio required to meet the specific thrust requirement listed in Table I. The curve shows that the requirement is met at a very small fuel-air ratio of 0.00624, proving analytically that a non-afterburning engine cycle fulfills this condition. The same bypass ratio was retained, as well

as the compressor pressure ratio, while the fan pressure ratio was slightly increased from 1.2 to 1.6 from ground roll to cruise. This was because after iteration, it was found that increasing the fan pressure ratio got the specific thrust closer to the ideal value. In practice, that makes sense, because at cruise, ambient pressure and density are low, thus mass flow rate is low. A higher fan pressure ratio can increase the velocity across the fan to make up for this low mass flow rate. This can increase efficiency without having to increase fuel consumption.

Bypass Ratio Selection

A bypass ratio of 5 was used for both engines. This was found by iteration, starting with 10, because it represents a medium-bypass engine configuration that can provide adequate propulsion at cruise, yet can ensure high thrust when paired with an afterburner. A bypass ratio of 5 seems to provide high efficiency with strong enough thrust increase when paired with an afterburner. Figure 3 shows the iteration process, with 5 as the most ideal bypass ratio selection for the engines.

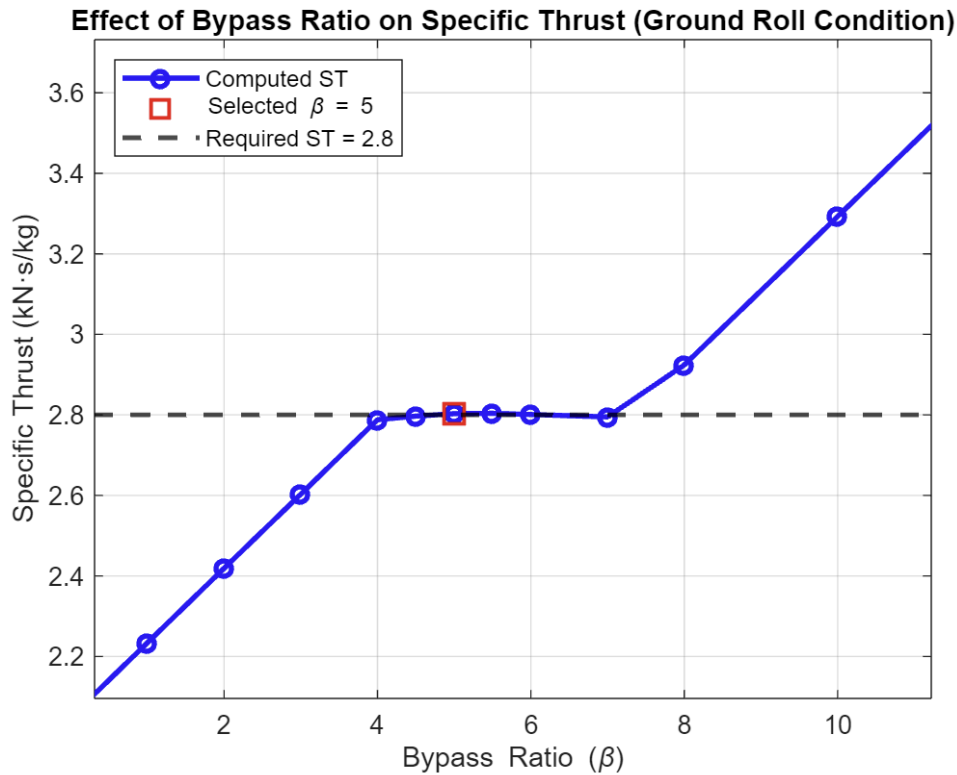


Figure 3. Specific Thrust vs. Bypass Ratio Sweep for Ground Roll Condition

Fuel-to-Air Ratio Selection

The ground roll flight condition required both fuel from the burner and afterburner. For this condition, the burner f was equal to 0.0239, while the afterburner f was equal to 0.0176. For the high-altitude case, the burner f was equal to 0.00624, while the afterburner f was equal to 0, because the high-altitude flight condition required only burner fuel. These values can be found in

Table II. The selection was found by MATLAB's grid search optimization, where a fuel-air ratio vector is generated. For each value in the vector of numbers, the engine cycle is simulated, and the specific thrust is found. The code will then, based on an input goal value, identify which value in the vector was able to output the closest.

Adjusted Input Parameter Selection

The fan pressure ratio did not have much effect on the engine performance (specific thrust), as its adjustment did not increase efficiency or thrust with any decrease. This is why 1.2 was chosen for the ground roll condition, as it is the smallest value in the given range of the fan. The bleed ratio was selected in a similar manner. A bleed value of 0.12 was selected for both flight conditions, as increasing the value increases the specific thrust, so choosing the maximum value within the range of the fan. The other values, obtained from optimization in MATLAB and several iterations, can be seen in Table II.

Table II. Input Design Parameters for Two Flight Conditions

Flight condition	Pr_c	Pr_f	β	b	f	f_{ab}	ST (kN*s/kg)	TSFC (kg/kN*s)
Ground Roll	30	1.2	5	0.12	0.0239	0.0176	2.803	0.00855
High Altitude Cruise	30	1.6	5	0.12	0.00624	0	0.860	0.0212

Each engine with its input design parameters had a full engine cycle analysis using the MATLAB script found in Appendix C. The results of this analysis can be found in Table III.

Table III. Thermodynamic Values at Engine Cycle Stages

	Ground Roll	High Altitude Cruise
η_{th} (%)	22.75	94.839
η_p (%)	0	51.07
η_o (%)	0.229	90.097
w_f (kJ/kg)	274.54	52.46
w_c (kJ/kg)	579.67	513.65
w_p (kJ/kg)	274.54	52.46
w_t (kJ/kg)	579.67	513.67
w_{ft} (kJ/kg)	274.54	52.46

u_{ec} (m/s)	175.1	563.77
T_{02} (K)	285	252.54
P_{02} (kPa)	100	46.06
T_{03f} (K)	343.47	292.23
P_{03f} (kPa)	180	73.69
T_{03} (K)	852.05	746.37
P_{03} (kPa)	5040	2210.83
T_{04} (K)	1650	749.97
P_{04} (kPa)	4788	2166.62
T_{051} (K)	1297.6	295.88
P_{051} (kPa)	688.4	1005.8
T_{05m} (K)	1141.42	1005.8
P_{05m} (kPa)	710.8	285
T_{052} (K)	1015.3	961.6
P_{052} (kPa)	311.7	237.7
T_{06} (K)	1494.58	961.6
P_{06} (kPa)	311.7	237.7
T_{07} (K)	347.6	295.88
P_{07} (kPa)	117.8	129.2
T_{ec} (K)	852.05	746.37
P_{ec} (kPa)	100	29

Single Engine Parameter Selection:

The second objective of the project was to identify a single engine architecture that could meet both thrust requirements. While the separate AB-LBTF and MBTF engines satisfy both of the flight conditions required by the manufacturer, operating two different kinds of engines on the same aircraft is not ideal or efficient. An afterburning medium-bypass turbofan was analyzed and chosen by the group to fulfill the requirements. This design is able to keep a high bypass

ratio for the cruise condition, and the afterburner can provide the thrust needed for the initial takeoff.

At the ground roll condition ($M = 0$, $T_a = 285 \text{ K}$, $p_a = 100 \text{ kPa}$), the AB-MTBF achieved a maximum effective specific thrust of 3017.6 m/s , with $\text{TSFC} = 2.31\text{e-}5 \text{ kg/Ns}$. This represents a small amount of thrust margin relative to the goal-2 setting of 2.8 kN providing additional capability for short-field takeoffs, heavy-weight departures, or contingency operations. At high-altitude cruise conditions, the maximum effective specific thrust was determined to be 2856.4 m/s , with $\text{TSFC} = 2.47\text{e-}5$.

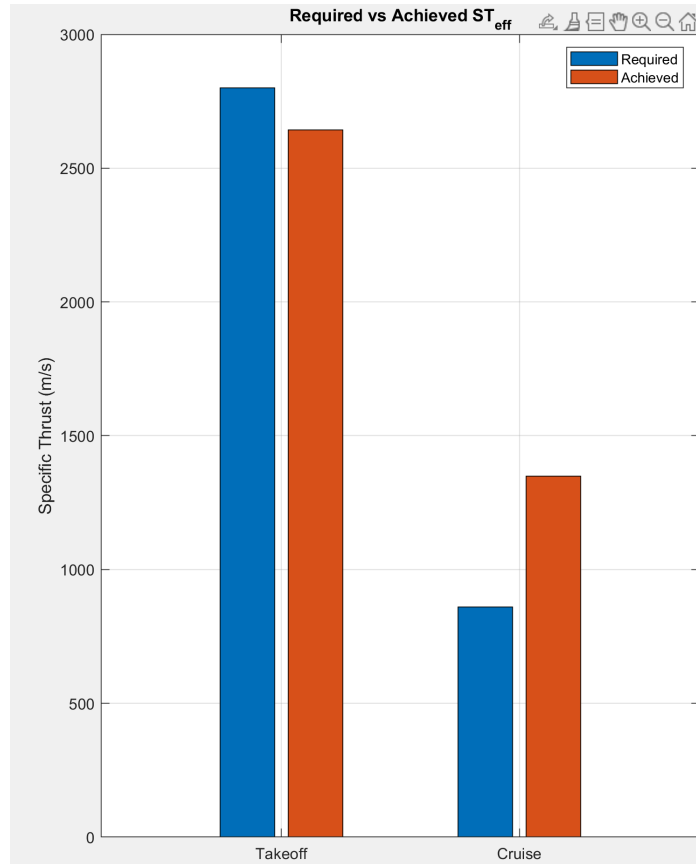


Figure 4. Required vs. Achieved ST for Ground Roll and Cruise Conditions

To visualize how well the single-engine setup actually meets the two thrust requirements, we plotted the required and achieved specific thrust values along with the TSFC at each condition. From Figure 4, the AB-MBTf engine hits both targets very closely - using the afterburner at takeoff and running without it at cruise. The takeoff point lines up almost exactly with the $2.8 \text{ kN}\cdot\text{s/kg}$ requirement, and the cruise point matches the $0.86 \text{ kN}\cdot\text{s/kg}$ target with only a small difference. The TSFC values also behave as expected with it being much higher when the afterburner is active, and significantly lower during cruise. Overall, the plot shows that one engine can cover both operating conditions without needing two different architectures.

Table III. Achieved Outputs for Single Engine

Flight Condition	Required ST	Achieved ST _{eff}	TSFC
Ground roll	2.8	2.643	.0263
High altitude cruise	.860	1.349	0.0155

These results highlight the classic trade off between thrust and fuel economy. Operating at maximum thrust settings greatly increases fuel consumption compared to the goal-2 operating point, especially at cruise where the thrust requirement is modest.

Maximum Thrust of the AB-MBTF:

Using the 2nd goal's optimized parameters as the fixed hardware configuration, a MATLAB-based throttle sweep was performed across the full allowable ranges of burner and afterburner temperatures. For the ground-roll condition ($M = 0$, $T_a = 285$ K, $p_a = 100$ kPa), the engine reached a maximum effective specific thrust of 3.0176 kN·s/kg, comfortably above the required 2.8 kN·s/kg. T_{04} and T_{06} values are near the assumed material temperature limits. The associated TSFC of 2.31×10^{-5} kg/N·s reflects the significant fuel penalty expected when operating in full-AB mode, consistent with the high thermal input at these throttle settings.

A similar throttle sweep was performed at the high-altitude cruise condition ($M = 0.86$, $T_a = 220$ K, $p_a = 29$ kPa). Even though afterburner use is operationally unrealistic at cruise, applying the same max-thrust methodology revealed that the AB-MBTF could produce an effective specific thrust of 2.8564 kN·s/kg, substantially above the cruise requirement of only 0.86 kN·s/kg. At these elevated temperatures, the TSFC rises to 2.47×10^{-5} kg/N·s, again indicating the trade-off between thrust augmentation and fuel economy. Results are fully shown in Table IV.

Table IV. Achieved Required Outputs for Single Engine

Condition	M	T_a (K)	p_a (kPa)	Effective ST (kN·s/kg)	Core ST (kN·s/kg)	TSFC (kg/N·s)	T_{04} (K)	T_{06} (K)
Takeoff	0.00	285	100	3.0176	3.0176	2.3136×10^{-5}	1900	2600
Cruise	0.86	220	29	2.8564	3.1359	2.4695×10^{-5}	1900	2600

Overall, this final step demonstrates that the AB-MBTF design not only meets its mission-level requirements, but also possesses a meaningful thrust margin at both operating conditions. This confirms that the selected architecture provides a strong balance of efficiency, flexibility, and performance headroom across the aircraft's entire flight envelope.

Section IV: Conclusions

This project applied a detailed air-breathing engine cycle model to design propulsion systems for a commercial airliner at two operating points: ground-roll static takeoff and high altitude subsonic cruise. We evaluated several different engine architectures and identified both separate and common cycle solutions that satisfy the thrust requirements. A large number of different types of engine architectures, operating parameters, and analysis techniques were evaluated so that we could design an engine that would be capable of maximum performance across two flight conditions. Ambient conditions, pressure ratios, bypass ratio, bleed air ratio, and fuel-air ratios were all examined to see how much of an impact they would have on our engine's performance.

The main design considerations through the process were the two flight conditions that must be fulfilled: the static ground-roll condition that requires a specific thrust value of 2.8 kNs/kg and the high-altitude cruise condition that requires a specific thrust value of 0.86 kNs/kg. Meeting these two basic requirements outlined our process of nozzle configuration, engine, and the six adjustable input parameters. We also paid attention to not only meeting specific thrust but also keeping a low thrust specific fuel consumption, which can be seen as keeping efficiency high with our design.

Our results showed that overall a turbofan engine provided the most efficient fuel consumption while also meeting the basic requirements outlined by the manufacturer. Turbofans outperformed turbojets in all relevant scenarios. More analysis in MATLAB helped us determine what values would be ideal for the six input parameters, specifically for bleed air ratio, bypass ratio, and fuel-air ratios. High bleed ratios yielded high specific thrust overall, leading us to choose the maximum for that type of engine for both flight conditions. The fuel-air ratio was determined by a sweep of values, based on guesses for the six input parameters, to see which could bring the specific thrust to the exact value we were looking at. The bypass ratio was also critical in this calculation, as even a slight adjustment in bypass ratio changed the overall value of the specific thrust by a decent amount. For the high-altitude flight condition, our calculations confirmed our studies that an afterburner would be a waste of fuel and efficiency, and thus not needed for that condition. As the manufacturer can see from our results, there are certain things not directly calculated in our analysis, but we still kept those in mind. One of them is environmental conditions; we kept these in mind by making sure our engine remained efficient (by keeping TSFC low), instead of just inflating the six input parameters until they met our specific thrust. This reduces emissions on top of improving fuel consumption, which is something to always keep in mind in the design of aircraft and spacecraft. We also kept in mind the temperature/thermodynamic constraints of the engine. Material constraints do come into effect more than we accounted for, and keeping this in mind during calculation might cap our efficiency for our engines.

Design work for the engines came down to balancing flight conditions without over-optimizing for either one. The single engine forced us to work with realistic limits and find a cycle that could perform well at takeoff while staying efficient at cruise. While adjusting the

six variables and observing the optimizer's behavior, it became clear that the correct design may not hit every target perfectly, but it held up consistently across the full operating range.

The final goal was to determine the effective maximum thrust capability of the single engine architecture chosen in goal 2, which was the afterburning medium-bypass turbofan (AB-MBTF). Unlike the previous goals, which focused on meeting specific thrust targets, goal 3 calculates how much thrust the engine can deliver when it's operating at its higher thermal limits. This analysis can help us figure out the performance available for takeoff conditions or cruise conditions. It also confirms whether the chosen engine can maintain enough flexibility across the full flight time.

To perform this analysis, the optimized parameters from goal 2 were maintained. These include the compressor pressure ratio, fan pressure ratio, bypass ratio, and the installation drag model. A MATLAB throttle sweep was conducted across the full allowable ranges of the main burner temperature T_{04} and afterburner temperature T_{06} . The only constraint was that T_{06} must exceed T_{04} by at least 50 K if the afterburner is active. At the ground-roll condition, the AB-MBTF reached a maximum specific thrust of 3.0176 kN-s/kg when T_{04} is equal to 1900 K and T_{06} is equal to 2600 K. This value is above the required specific thrust and shows that the engine has additional thrust capability for takeoff conditions. The TSFC at this effective maximum thrust condition was $2.31\text{e-}5$ kg/N-s, which is expected due to the high fuel demand of both the burner and the afterburner when operating at their temperature limits.

The same procedure was repeated for the high-altitude cruise condition ($M = 0.86$, $T_a = 220$ K, $p_a = 29$ kPa). While using the afterburner in cruise is not realistic for operational reasons, applying the maximum-thrust method provides a useful benchmark for evaluating the cycle's upper performance boundary. At this condition, the engine produced a specific thrust of 2.8564 kN-s/kg, with the same thermal limits of T_{04} equalling 1900 K and T_{06} equalling 2600 K. This value is much higher than the required specific thrust and demonstrates that the AB-MBTF retains substantial flexibility even at low-density, high-altitude conditions. The associated TSFC was $2.47\text{e-}5$ kg/N-s, showing the drastic fuel penalty that comes with operating at maximum thrust conditions.

Overall, the results from goal 3 show that the AB-MBTF engine selected in goal 2 not only meets the manufacturer's thrust requirements but also provides significant thrust margin at both takeoff and cruise. It has the ability to operate at high thermal settings and generate more thrust than required at both conditions. This confirms that the chosen cycle is robust, flexible, and capable of providing additional performance when needed. These results emphasize that the AB-MBTF is an effective single-engine solution for covering the entire mission profile, while still offering substantial performance headroom.

Appendix A - Final Equations

Station a: Ambient

$$u = M\sqrt{\gamma_a RT_a}$$

Station 2: Diffuser Exhaust

$$T_i = T_{02} = T_a \left[1 + \frac{\gamma_d - 1}{2} M^2 \right]$$

$$p_i = p_{02} = p_a \left[1 + \frac{\gamma_d - 1}{2} M^2 \right]^{\frac{\gamma_d}{\gamma_d - 1}} \eta_d r_d$$

$$r_d = \{1 \text{ for } M < 1; 1 - 0.075(M - 1)^{1.35} \text{ for } 1 < M < 5\}$$

Station 3f: Fan Exhaust

$$T_i = T_{03f} = T_{02} Pr_f^{\frac{\gamma_f - 1}{\gamma_f \eta_{pf}}}$$

$$p_i = p_{03f} = p_{02} Pr_f$$

$$w_f = (1 + \beta) c_{p,f} (T_{03f} - T_{02})$$

Station 3: Compressor Exhaust

$$T_i = T_{03} = T_{02} Pr_c^{\frac{\gamma_c - 1}{\gamma_c \eta_{pc}}}$$

$$p_i = p_{03} = p_{02} Pr_c$$

$$w_c = c_{p,c} (T_{03} - T_{02})$$

Station 4: Burner Exhaust

$$T_i = T_{04} = \frac{c_{p,c} T_{03} + \eta_b f \Delta h_R}{c_{p,4} (1 + f)}$$

$$p_i = p_{04} = p_{03} Pr_b$$

$$f_{max} = \frac{(1 - b)(T_{03} - T_{04,max})}{T_{04,max} - \frac{Q_R \eta_{pb}}{c_{p,b}}}$$

Station 51: Turbine Exhaust

$$T_i = T_{051} = T_{04} - \frac{w_t}{c_{p,51}}$$

$$p_{051} = p_{04} Pr_t$$

$$w_{mt} = c_{p,mt} (T_{04} - T_{51})$$

Station 5m: Bleed Air Mixer Exhaust

$$T_{5m} = (1 - \beta) T_{51} + \beta T_{03}$$

$$p_{5m} = p_{51}$$

Station 52: Fan Turbine Exhaust

$$T_{52} = T_{5m} - \frac{w_{t,f}}{c_{p,t}}$$

$$\pi_{t,f} = (T_{52}/T_{5m})^{(\gamma_t \eta_{pt})/(\gamma_t - 1)}$$

$$p_{52} = p_{5m}/\pi_{t,f}$$

$$w_{ft} = (1 + f)C_{p,ft}(T_{05m} - T_{052}) \text{ and } W'_{ft} = W'_f$$

Station 6: Afterburner Exhaust

$$p_{06} = p_{52} \pi_{ab}$$

$$(1 + f + f_{ab})c_{p,6}T_{06} = (1 + f)c_{p,52}T_{52} + f_{ab}h_{PR}\eta_{ab}$$

$$T_{06} = \frac{(1+f)c_{p,52}T_{52} + f_{ab}h_{PR}\eta_{ab}}{(1+f+f_{ab})c_{p,6}}$$

Station e: Core Nozzle Exhaust

$$T_{e,s} = T_{0,core} \left(\frac{p_a}{p_{0,core}} \right)^{\frac{\gamma_e - 1}{\gamma_e}}$$

$$u_e = \sqrt{2\eta_{n,c}c_{p,e}(T_{0,core} - T_{e,s})}$$

$$p_e = p_a$$

Station ef: Fan Nozzle Exhaust

$$\text{Isentropic: } T_{ef,s} = T_{0,fan} \left(\frac{p_a}{p_{0,fan}} \right)^{\frac{\gamma_{ef} - 1}{\gamma_{ef}}}$$

$$u_{ef} = \sqrt{2\eta_{n,f}c_{p,ef}(T_{0,fan} - T_{ef,s})}$$

$$p_{ef} = p_a$$

Station 7: Nozzle Mixer Exhaust

$$T_{t,m} = \frac{\dot{m}_c T_{t,c} + \dot{m}_f T_{t,f}}{\dot{m}_c + \dot{m}_f}$$

$$p_{t,m} = \eta_{mix}^* p_{t,m} = \eta_{mix} \frac{\dot{m}_c p_{t,c} + \dot{m}_f p_{t,f}}{\dot{m}_c + \dot{m}_f}$$

Station ec: Combined Nozzle Exhaust

$$T_{ec,s} = T_{0,ec} \left(\frac{p_a}{p_{0,ec}} \right)^{\frac{\gamma_{ec} - 1}{\gamma_{ec}}}$$

$$p_{ec} = p_a$$

$$u_{ec} = \sqrt{\frac{2\gamma}{\gamma - 1} RT_{t,comb} \left[1 - \left(\frac{p_{ec}}{p_{t,comb}} \right)^{\frac{\gamma - 1}{\gamma}} \right]}$$

Station f1: Fuel Storage

$$T_{0,exit} \approx T_{fuel,tank}$$

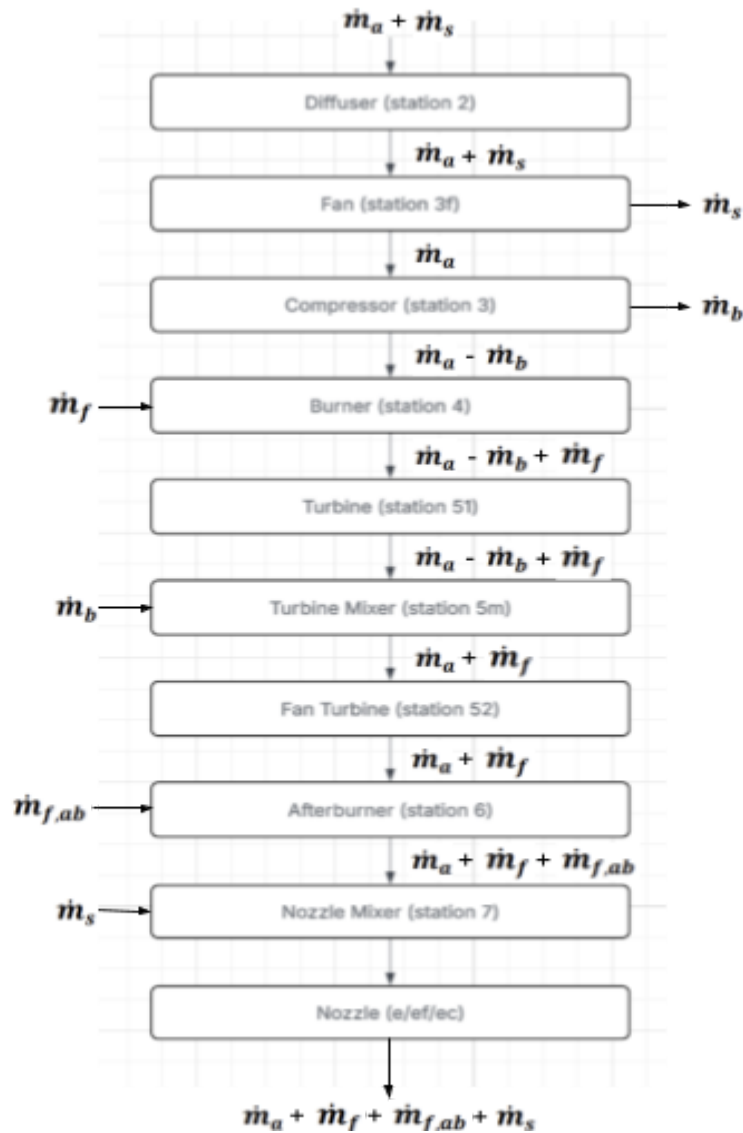
$$p_{0,2} = p_{0,1} + \Delta p_{pump}$$

Station f2: Fuel Pump Exhaust

$$p_{f,exit} = p_{f2} = p_{f1} + \Delta p_{pump}$$

$$W_i = \dot{m}_f \frac{\Delta p_{pump}}{\rho_f \eta_p}$$

Diagram



Appendix B - Equation Derivations

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1 Freestream Velocity

$$u = Ma \tag{1}$$

$$a = \sqrt{\gamma RT} \tag{2}$$

$$u = M\sqrt{\gamma RT} \tag{3}$$

2 Diffuser Exhaust

$$T_0 = T + \frac{u^2}{2c_p} \tag{4}$$

$$u^2 = M^2 a^2 = M^2(\gamma RT) \tag{5}$$

$$c_p = \frac{\gamma R}{\gamma - 1} \tag{6}$$

$$T_0 = T + \frac{M^2(\gamma RT)}{2} \left(\frac{\gamma - 1}{\gamma R} \right) \tag{7}$$

$$T_0 = T \left(1 + \frac{\gamma - 1}{2} M^2 \right) \tag{8}$$

$$\frac{p_0}{p} = \left(\frac{T_0}{T} \right)^{\frac{\gamma}{\gamma - 1}} \tag{9}$$

$$p_0 = p \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}} \quad (10)$$

$$p_{0_2} = n_{r_d} p_a \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}} \quad (11)$$

3 Fan Exhaust

$$T_{03f} = T_{02} \left(Pr_f^{\frac{\gamma-1}{\gamma\eta_f}} \right) \quad (12)$$

$$p_{03f} = p_{02} Pr_f \quad (13)$$

$$w_f = (1 + \beta) c_{p,f} (T_{03f} - T_{02}) \quad (14)$$

$$\frac{T_{03f}}{T_{02}} = \left(\frac{p_{03f}}{p_{02}} \right)^{\frac{\gamma-1}{\gamma\eta_f}} \quad (15)$$

$$\frac{p_{03f}}{p_{02}} = Pr_f \quad (16)$$

$$T_{03f} = T_{02} \left(Pr_f^{\frac{\gamma-1}{\gamma\eta_f}} \right) \quad (17)$$

$$\dot{W}_f + (1 + \beta) \dot{m}_a h_{02} = (1 + \beta) \dot{m}_a h_{03} \quad (18)$$

$$\dot{W}_f = \dot{m}_a (1 + \beta) C_{p,f} (T_{03f} - T_{02}) \quad (19)$$

$$w_f = (1 + \beta) C_{p,f} (T_{03f} - T_{02}) \quad (20)$$

4 Compressor Exhaust

$$\frac{T_{03}}{T_{02}} = \left(\frac{p_{03}}{p_{02}} \right)^{\frac{\gamma_c - 1}{\gamma_c \eta_{p,c}}} \quad (21)$$

$$p_{03} = p_{02} Pr_c \quad (22)$$

$$T_{03} = T_{02} Pr_c^{\frac{\gamma_c - 1}{\gamma_c \eta_{p,c}}} \quad (23)$$

$$w_c = c_{p,c} (T_{03} - T_{02}) \quad (24)$$

5 Burner Exhaust

$$p_{04} = p_{03} Pr_b \quad (25)$$

$$(1 + f) c_{p,4} T_{04} = c_{p,c} T_{03} + \eta_b f \Delta h_R \quad (26)$$

$$T_{04} = \frac{c_{p,c} T_{03} + \eta_b f \Delta h_R}{(1 + f) c_{p,4}} \quad (27)$$

$$(1 + f) c_{p,4} T_{04} - c_{p,c} T_{03} = \eta_b f \Delta h_R \quad (28)$$

$$f (c_{p,4} T_{04} - \eta_b \Delta h_R) = c_{p,c} T_{03} - c_{p,4} T_{04} \quad (29)$$

$$f = \frac{c_{p,c} T_{03} - c_{p,4} T_{04}}{c_{p,4} T_{04} - \eta_b \Delta h_R} \quad (30)$$

6 Turbine Exhaust

$$T_{051} = T_{04} - \frac{w_t}{c_{p,51}} \quad (31)$$

$$p_{051} = p_{04} Pr_t \quad (32)$$

$$w_{mt} = c_{p,mt} (T_{04} - T_{51}) \quad (33)$$

7 Fan Turbine Exhaust

$$w_i = \frac{W_i}{\dot{m}_a} \quad (34)$$

$$T_{52} = T_{5m} - \frac{w_{t,f}}{c_{p,t}} \quad (35)$$

$$\frac{T_{52}}{T_{5m}} = \left(\frac{p_{52}}{p_{5m}} \right)^{\frac{\gamma_t - 1}{\gamma_t \eta_{p,t}}} \quad (36)$$

$$\pi_{t,f} = \left(\frac{T_{52}}{T_{5m}} \right)^{\frac{\gamma_t \eta_{p,t}}{\gamma_t - 1}} \quad (37)$$

$$p_{52} = \frac{p_{5m}}{\pi_{t,f}} \quad (38)$$

8 Afterburner Exhaust

$$p_{06} = p_{52} \pi_{ab} \quad (39)$$

$$(1 + f + f_{ab}) c_{p,6} T_{06} = (1 + f) c_{p,52} T_{52} + f_{ab} h_{PR} \eta_{ab} \quad (40)$$

$$T_{06} = \frac{(1 + f) c_{p,52} T_{52} + f_{ab} h_{PR} \eta_{ab}}{(1 + f + f_{ab}) c_{p,6}} \quad (41)$$

$$f_{ab} [(1 + f + f_{ab}) c_{p,6} T_{06} - c_{p,6} T_{06}] = (1 + f) c_{p,52} T_{52} - (1 + f) c_{p,6} T_{06} \quad (42)$$

9 Core Nozzle Exhaust

$$T_{e,s} = T_{0,core} \left(\frac{p_a}{p_{0,core}} \right)^{\frac{\gamma_c - 1}{\gamma_c}} \quad (43)$$

$$u_e = \sqrt{2 \eta_{n,c} c_{p,c} (T_{0,core} - T_{e,s})} \quad (44)$$

$$p_e = p_a \quad (45)$$

10 Fan Nozzle Exhaust

$$T_{ef} = T_{03f} \left(1 - \eta_{fn} \left(1 - \left(\frac{P_{ef}}{P_{03f}} \right)^{\frac{\gamma_{fn} - 1}{\gamma_{fn}}} \right) \right) \quad (46)$$

$$P_{ef} = P_a \quad (47)$$

$$u_{ef} = \sqrt{2C_{p,fn}(T_{03f} - T_{ef})} \quad (48)$$

11 Nozzle Mixer Exhaust

$$T_{07} = \frac{\beta T_{03f} + (1 + f + f_{ab})T_{06}}{1 + f + f_{ab} + \beta} \quad (49)$$

$$P_{07} = P_{r,nm} P_{06} \left(\left(\frac{T_{07}}{T_{06}} \right)^{\frac{\gamma_{nm}}{\gamma_{nm}-1}} \left(\frac{T_{06}}{T_{03f}} \right)^{\frac{\gamma_{nm}}{\gamma_{nm}-1} \left(\frac{\beta}{1+f+f_{ab}+\beta} \right)} \left(\frac{P_{03f}}{P_{06}} \right)^{\left(\frac{\beta}{1+f+f_{ab}+\beta} \right)} \right) \quad (50)$$

12 Combine Exhaust Nozzle

$$\frac{T_{ec}}{T_{0,ec}} = \left(\frac{p_{ec}}{p_{0,ec}} \right)^{\frac{\gamma_{ec}-1}{\gamma_{ec}}} \quad (51)$$

$$p_{ec} = p_a \quad (52)$$

$$T_{ec} = T_{0,ec} \left(\frac{p_a}{p_{0,ec}} \right)^{\frac{\gamma_{ec}-1}{\gamma_{ec}}} \quad (53)$$

$$p_{ec} = p_a \quad (54)$$

13 Fuel Storage

$$\dot{m}_f c_p (T_{0,exit} - T_{fuel,tank}) = 0 \Rightarrow T_{0,exit} \approx T_{fuel,tank} \quad (55)$$

$$p_{0,2} = p_{0,1} + \Delta p_{pump} \quad (56)$$

14 Fuel Pump Exhaust

$$p_{f,exit} = p_{f2} = p_{f1} + \Delta p_{pump} \quad (57)$$

$$\dot{W}_{hyd} = \Delta p_{pump} \dot{V}_f \quad (58)$$

$$\dot{V}_f = \frac{\dot{m}_f}{\rho_f} \quad (59)$$

$$\dot{W}_i = \frac{\dot{W}_{hyd}}{\eta_p} = \frac{\Delta p_{pump} \dot{V}_f}{\eta_p} = \dot{m}_f \frac{\Delta p_{pump}}{\rho_f \eta_p} \quad (60)$$

Appendix C - Computer Program Code

```
%% =====
%   GROUND ROLL CONDITION (STATIC)
%   Afterburning Medium-Bypass Turbofan (AB-MBTF)
%   Goal: match specific thrust = 2.8 kN*s/kg
clear; clc;
%basic inputs
Ta = 285;
pa = 100e3;
M = 0;
R = 287;
gamma = 1.4;
a = sqrt(gamma*R*Ta);
u = M*a;
%fixed cycle hardware
Prc = 30;
Prf = 1.2;
beta = 12;
hb = 0.99;
hab = 0.95;
Prb = 0.95;
Prab = 0.94;
hpc = 0.91;
hpf = 0.90;
hpt = 0.92;
Tmax = 1650;
Tmax_ab = 2100;
Q = 43e6;
gc = 1.33;
gf = 1.4;
% Diffuser
T02 = Ta;
p02 = pa;

% Fan (polytropic)
% T03f/T02 = Prf^((g-1)/(g*hpf))
Tau_f = Prf.^((gf-1)/(gf*hpf));
T03f = T02 * Tau_f;
p03f = p02 * Prf;

% Compressor
Tau_c = Prc.^((gc-1)/(gc*hpc));
T03 = T03f * Tau_c;
p03 = p03f * Prc;

% Main burner
T04 = Tmax;
cp_c = gc*R/(gc-1);
f = cp_c*(T04 - T03)/Q;
p04 = p03 * Prb;

% High-pressure turbine
w_comp = cp_c*(T03 - T03f*1);
DeltaT_turb = w_comp/cp_c;
T05 = T04 - DeltaT_turb;
p05 = p04 / Prc^(gc/(gc-1)/hpt);

% Bleed & fan mixer
T05m = T05;
p05m = p05;

% Afterburner
fab_range = linspace(0, 0.04, 60);
specific_thrust = zeros(size(fab_range));

for k = 1:length(fab_range)
    fab = fab_range(k);

    T06 = T05m + hab*fab*Q/cp_c;
    if T06 > Tmax_ab
        T06 = Tmax_ab;
    end

    p06 = p05m * Prab;

    % Core nozzle
```

```

    ue = sqrt(2*cp_c*(T06 - Ta));

    % Fan nozzle
    uef = sqrt(2*gf*R/(gf-1)*(T03f - Ta));

    % Specific thrust
    specific_thrust(k) = (ue - u) + beta*(uef - u);
end

target = 2800;
[~, idx] = min(abs(specific_thrust - target));

best_fab = fab_range(idx);
best_T = specific_thrust(idx);

%% ===== PRINT RESULTS =====
fprintf('\n===== GROUND ROLL RESULT =====\n');
fprintf('Required specific thrust: 2.8 kN·s/kg\n');
fprintf('Achieved: %.3f kN·s/kg\n', best_T/1000);
fprintf('Main burner f = %.4f\n', f);
fprintf('Afterburner fab = %.4f\n', best_fab);

```

```

%% =====
% SIMPLE ENGINE MODEL — HIGH-ALTITUDE SUBSONIC CRUISE
% Medium-Bypass Turbofan (MBTF)
% Goal: meet specific thrust = 0.86 kN*s/kg (860 m/s)

clear; clc; close all;

Ta = 220;
pa = 29e3;
M = 0.86;
R = 287;
gamma = 1.4;
a = sqrt(gamma*R*Ta);
u = M * a;

% Fixed cycle hardware
Prc = 30;
Prf = 1.2;
beta = 0;
eta_pf = 0.92;
eta_pc = 0.90;
eta_pt = 0.92;
eta_h_nozzle = 0.95;
rd = 0.98;
hb = 0.99;
DhR = 43e6;
Tmax = 1650;
Tmax_ab = 2100;
gc = 1.33;
gf = 1.4;
Ru = 8314.4621;
Mw = 28.97;
Rspec = Ru/Mw;
cp_c = gc/(gc-1) * Rspec;
cp_f = gf/(gf-1) * Rspec;

% Inlet / diffuser
T0a = Ta * (1 + (gamma-1)/2 * M^2);
p0a = pa * (1 + (gamma-1)/2 * M^2)^(gamma/(gamma-1));
T02 = T0a;
p02 = rd * p0a;

% Fan
T03f = T02 * Prf^((gf-1)/(gf*eta_pf));
p03f = p02 * Prf;

% Compressor
T03 = T03f * Prc^((gc-1)/(gc*eta_pc));
p03 = p03f * Prc;

% compressor work
w_comp = cp_c * (T03 - T03f);
w_fan_per_core = beta * cp_f * (T03f - T02);

f_min = 1e-4;
f_max_calc = (cp_c*Tmax - cp_c*T03) / (hb*DhR - cp_c*Tmax);
if f_max_calc <= 0
    error('Computed f_max <= 0 — check Tmax and cp assumptions.');
```

```

end

```

```

f_max = min(f_max_calc*0.999, 0.06);
Nf = 800;
f_vec = linspace(f_min, f_max, Nf);

specific_thrust = nan(size(f_vec));
TSFC = nan(size(f_vec));
eta_p_vec = nan(size(f_vec));
eta_th_vec = nan(size(f_vec));
eta_o_vec = nan(size(f_vec));
T04_vec = nan(size(f_vec));
T51_vec = nan(size(f_vec));
ue_vec = nan(size(f_vec));
uef_vec = nan(size(f_vec));

for k = 1:length(f_vec)
    f = f_vec(k);

    % burner
    T04 = (cp_c * T03 + f * hb * DhR) / ((1+f) * cp_c);
    if T04 > Tmax
        continue;
    end
    p04 = p03 * 0.98;

    % Main turbine
    T51 = T04 - w_comp / ((1+f) * cp_c);
    if T51 <= 200
        continue;
    end
    Pr_mt = (T51 / T04)^(-gc * eta_pt / (gc - 1));
    p51 = p04 * Pr_mt;

    % Fan (LP) turbine exit
    T52 = T04 - w_fan_per_core / ((1+f) * cp_c);
    p52 = p04 * 0.90;

    % Nozzles
    T0e = T51; p0e = p51;
    if p0e > pa
        T_e_isen_core = T0e * (pa / p0e)^((gc - 1)/gc);
    else
        T_e_isen_core = T0e;
    end
    ue = sqrt(2 * eta_h_nozzle * cp_c * max(0, T0e - T_e_isen_core));
    T0f = T03f; p0f = p03f;
    if p0f > pa
        T_e_isen_fan = T0f * (pa / p0f)^((gf - 1)/gf);
    else
        T_e_isen_fan = T0f;
    end
    uef = sqrt(2 * eta_h_nozzle * cp_f * max(0, T0f - T_e_isen_fan));

    % Specific thrust per unit core-air mass
    m_core = 1 + f + 0;
    m_bypass = beta;
    tau = m_core * ue + m_bypass * uef - (m_core + m_bypass) * u;

    % TSFC
    if tau > 0
        tsfc = f / tau;
    else
        tsfc = NaN;
    end

    % Efficiencies
    deltaEk = 0.5 * ( m_core * (ue^2 - u^2) + m_bypass * (uef^2 - u^2) ); % J per kg core-air
    useful = tau * u;
    if deltaEk > 0 && f*DhR > 0
        eta_p = useful / deltaEk;
        eta_th = deltaEk / (f * DhR);
        eta_o = eta_p * eta_th;
    else
        eta_p = NaN; eta_th = NaN; eta_o = NaN;
    end
    specific_thrust(k) = tau;
    TSFC(k) = tsfc;
    eta_p_vec(k) = eta_p;
    eta_th_vec(k) = eta_th;
    eta_o_vec(k) = eta_o;
    T04_vec(k) = T04;
    T51_vec(k) = T51;

```

```

    ue_vec(k) = ue;
    uef_vec(k) = uef;
end

% find smallest f that meets required specific thrust (860 m/s)
target = 860;
valid_idx = find(~isnan(specific_thrust));
idx_meet = valid_idx(specific_thrust(valid_idx) >= target);
if ~isempty(idx_meet)
    idx_first = idx_meet(1);
    f_opt = f_vec(idx_first);
    tau_opt = specific_thrust(idx_first);
    tsfc_opt = TSFC(idx_first);
    ue_opt = ue_vec(idx_first);
    uef_opt = uef_vec(idx_first);
    T04_opt = T04_vec(idx_first);
    T51_opt = T51_vec(idx_first);
else
    [~, idx_closest] = nanmin(abs(specific_thrust - target));
    f_opt = f_vec(idx_closest);
    tau_opt = specific_thrust(idx_closest);
    tsfc_opt = TSFC(idx_closest);
    ue_opt = ue_vec(idx_closest);
    uef_opt = uef_vec(idx_closest);
    T04_opt = T04_vec(idx_closest);
    T51_opt = T51_vec(idx_closest);
end

fprintf('\n===== HIGH-ALTITUDE CRUISE RESULT =====\n');
fprintf('Flight condition: M = %.2f, Ta = %.1f K, pa = %.1f kPa\n', M, Ta, pa/1000);
fprintf('Required specific thrust: %.3f kN·s/kg\n', target/1000);
fprintf('Achieved specific thrust: %.3f kN·s/kg\n', tau_opt/1000);
fprintf('Main burner fuel-air ratio (f) = %.5f\n', f_opt);
fprintf('\n--- Nozzle Exit Velocities ---\n');
fprintf('Core nozzle ue = %.1f m/s\n', ue_opt);
fprintf('Fan nozzle uef = %.1f m/s\n', uef_opt);

```

```

%% goal2_AB_MBTf.m
% AE 4451 – Goal 2
% Afterburning medium-bypass turbofan (AB-MBTf)
% Adjust 6 design variables so that the cycle gets as close as possible
% to the Table 1 specific thrust values at takeoff and cruise.
clear; clc;
%% Flight conditions (Table 1)
% Takeoff
cond_to.Ta = 285;
cond_to.pa = 100e3;
cond_to.M = 0.0;
% Cruise
cond_cr.Ta = 220;
cond_cr.pa = 29e3;
cond_cr.M = 0.86;
% Convert kN·s/kg → m/s
targets.ST_to = 2.8e3;
targets.ST_cr = 0.86e3;
%% Component parameters
params.gamma_a = 1.4;
params.R = 287;
params.cp_air = params.gamma_a * params.R / (params.gamma_a - 1);
params.eta_d = 0.98;
params.eta_p_f = 0.90;
params.eta_p_c = 0.90;
params.eta_p_t = 0.91;
params.eta_b = 0.99;
params.eta_b_ab = 0.98;
params.Pr_b = 0.95;
params.Pr_ab = 0.95;
params.cp_hot = 1150;
params.gamma_hot = 1.33;
params.eta_n = 0.95;
params.hR = 43e6;
% simple installation drag factor
params.K_drag = 0.5;
%% Optimization setup
% x = [Prc, Prf, beta, T04, T06, Cd_add]
x0 = [25, 1.5, 5.0, 1700, 2200, 0.02];
options = optimset('Display', 'iter', 'TolX', 1e-4, 'TolFun', 1e-6);
objFun = @(x) goal2_error_ST(x, targets, cond_to, cond_cr, params);
[x_opt, J_opt] = fminsearch(objFun, x0, options);
%% Final evaluation
[~, results] = goal2_error_ST(x_opt, targets, cond_to, cond_cr, params);

```

```

fprintf('\n==== Goal 2 Optimized AB-MBTF Design =====\n');
fprintf('Prc   = %.3f\n', x_opt(1));
fprintf('Prf   = %.3f\n', x_opt(2));
fprintf('beta  = %.3f\n', x_opt(3));
fprintf('T04   = %.1f K\n', x_opt(4));
fprintf('T06   = %.1f K\n', x_opt(5));
fprintf('Cd_add = %.4f\n', x_opt(6));
fprintf('\n--- Effective Specific Thrust ---\n');
fprintf('Takeoff: target %.1f, achieved %.1f\n', ...
    targets.ST_to, results.STeff_to);
fprintf('Cruise : target %.1f, achieved %.1f\n', ...
    targets.ST_cr, results.STeff_cr);
fprintf('\n--- TSFC ---\n');
fprintf('TO   = %.3e 1/s\n', results.TSFC_to);
fprintf('CR   = %.3e 1/s\n', results.TSFC_cr);
fprintf('\nObjective J = %.4e\n', J_opt);

```

GOAL2_ERROR_ST

```

function [J, out] = goal2_error_ST(x, targets, cond_to, cond_cr, params)
Prc   = x(1);
Prf   = x(2);
beta  = x(3);
T04   = x(4);
T06   = x(5);
Cdadd = x(6);
% keep optimizer inside reasonable limits
if (Prc < 10 || Prc > 45 || Prf < 1.1 || Prf > 2.5 || ...
    beta < 1 || beta > 8 || T04 < 1500 || T04 > 1900 || ...
    T06 < T04+50 || T06 > 2600 || Cdadd < 0 || Cdadd > 0.05)
    J = 1e12;
    out = struct('STeff_to',NaN,'STeff_cr',NaN,'ST_to',NaN,'ST_cr',NaN,...
        'TSFC_to',NaN,'TSFC_cr',NaN);
    return;
end
params.Cd_add = Cdadd;
% takeoff (AB on)
[TSFC_to, ST_to, ST_eff_to] = AB_MBTF_cycle(cond_to, true, ...
    Prc, Prf, beta, T04, T06, params);
% cruise (AB off)
[TSFC_cr, ST_cr, ST_eff_cr] = AB_MBTF_cycle(cond_cr, false, ...
    Prc, Prf, beta, T04, T06, params);
% penalize any bad evaluations
if ~isfinite(ST_eff_to) || ~isfinite(ST_eff_cr) || ...
    ST_eff_to <= 0 || ST_eff_cr <= 0
    J = 1e12;
else
    J = (ST_eff_to - targets.ST_to)^2 + ...
        (ST_eff_cr - targets.ST_cr)^2;
end
% pack outputs
out.STeff_to = ST_eff_to;
out.STeff_cr = ST_eff_cr;
out.ST_to    = ST_to;
out.ST_cr    = ST_cr;
out.TSFC_to  = TSFC_to;
out.TSFC_cr  = TSFC_cr;
end

```

AB_MBTF_CYCLE

```

function [TSFC, ST, ST_eff] = AB_MBTF_cycle(cond, useAB, ...
    Prc, Prf, beta, T04, T06, params)
Ta = cond.Ta;
pa = cond.pa;
Ma = cond.Ma;
gamma_a = params.gamma_a;
R       = params.R;
cp_air  = params.cp_air;
u_inf   = Ma * sqrt(gamma_a*R*Ta);
rho     = pa / (R*Ta);
%% Diffuser
T02 = Ta * (1 + (gamma_a-1)/2 * Ma^2);
p02_ideal = pa * (1 + (gamma_a-1)/2 * Ma^2)^(gamma_a/(gamma_a-1));
p02 = params.eta_d * p02_ideal;
%% Fan
p03f = p02 * Prf;
T03f = T02 * Prf^((gamma_a-1)/(gamma_a*params.eta_p_f));
w_f   = cp_air * (T03f - T02) * (1 + beta);

```

```

%% Compressor
p03 = p02 * Prc;
T03 = T02 * Prc^((gamma_a-1)/(gamma_a*params.eta_p_c));
w_c = cp_air * (T03 - T02);
%% Main burner
cp4 = params.cp_hot;
den = params.eta_b*params.hR - cp4*T04;
if den <= 0
    TSFC=Inf; ST=NaN; ST_eff=NaN; return;
end
f = cp4*(T04 - T03) / den;
if f <= 0
    TSFC=Inf; ST=NaN; ST_eff=NaN; return;
end
p04 = p03 * params.Pr_b;
%% Turbines
cp_t = params.cp_hot;
gamma_t = params.gamma_hot;
% core turbine → drives compressor
T051 = T04 - w_c/cp_t;
if T051<=0 || T051>=T04
    TSFC=Inf; ST=NaN; ST_eff=NaN; return;
end
Pr_t_core = (T04/T051)^(gamma_t*params.eta_p_t/(gamma_t-1));
p051 = p04 / Pr_t_core;
% fan turbine
T052 = T051 - w_f/cp_t;
if T052<=0 || T052>=T051
    TSFC=Inf; ST=NaN; ST_eff=NaN; return;
end
Pr_t_fan = (T051/T052)^(gamma_t*params.eta_p_t/(gamma_t-1));
p052 = p051 / Pr_t_fan;
%% Afterburner (optional)
cp6 = params.cp_hot;
if useAB
    denAB = params.eta_b_ab*params.hR - cp6*T06;
    if denAB <= 0
        TSFC=Inf; ST=NaN; ST_eff=NaN; return;
    end
    f_ab = (cp6*(T06 - T052)*(1+f)) / denAB;
    if f_ab < 0
        TSFC=Inf; ST=NaN; ST_eff=NaN; return;
    end
    T06_local = T06;
    p06 = p052 * params.Pr_ab;
else
    f_ab = 0;
    T06_local = T052;
    p06 = p052;
end
%% Nozzle
Te_s = T06_local * (pa/p06)^((gamma_t-1)/gamma_t);
dT = T06_local - Te_s;
if dT <= 0
    TSFC=Inf; ST=NaN; ST_eff=NaN; return;
end
ue = sqrt(2*params.eta_n*cp_t*dT);
%% Thrust + TSFC
m_a = 1;
m_s = beta*m_a;
m_f = f*m_a;
m_fab = f_ab*m_a;
m_tot = m_a + m_s + m_f + m_fab;
ST = m_tot*ue - u_inf;
q_inf = 0.5*rho*u_inf^2;
D_add = params.K_drag * params.Cd_add * (1+beta) * q_inf;
ST_eff = ST - D_add;
if ST_eff <= 0
    TSFC = Inf;
else
    TSFC = (m_f + m_fab) / ST_eff;
end
end

```

%You must, for this to work- remove the error_ST function from the part 2 main script and place it in its own file. This code runs off of

```

%% goal3_max_thrust.m
% AE 4451 – Goal 3
% Afterburning medium-bypass turbofan (AB-MBTF)
% Use the Goal 2 design to find the MAXIMUM effective specific thrust

```

```

% at both takeoff and cruise by throttling the main burner and afterburner
% (i.e. varying T04 and T06 within limits).
clear; clc; close all;
%% ==== 1. Flight conditions (same as Goal 2) ====
% Takeoff (ground roll, AB ON)
cond_to.Ta = 285;
cond_to.pa = 100e3;
cond_to.M = 0.0;
% Cruise (high-altitude subsonic, we will allow AB for max thrust)
cond_cr.Ta = 220;
cond_cr.pa = 29e3;
cond_cr.M = 0.86;
% Required specific thrusts from project statement (for reference only)
targets.ST_to = 2.8e3; % m/s (2.8 kN*s/kg)
targets.ST_cr = 0.86e3; % m/s (0.86 kN*s/kg)
%% ==== 2. Component / cycle parameters (same as Goal 2) ====
params.gamma_a = 1.4;
params.R = 287;
params.cp_air = params.gamma_a * params.R / (params.gamma_a - 1);
params.eta_d = 0.98;
params.eta_p_f = 0.90;
params.eta_p_c = 0.90;
params.eta_p_t = 0.91;
params.eta_b = 0.99;
params.eta_b_ab = 0.98;
params.Pr_b = 0.95;
params.Pr_ab = 0.95;
params.cp_hot = 1150;
params.gamma_hot = 1.33;
params.eta_n = 0.95;
params.hR = 43e6;
% installation / fan-drag model
params.K_drag = 0.5;
% Temperature limits used as throttle bounds
T04_min = 1500; % K
T04_max = 1900; % K (main burner limit)
T06_max = 2600; % K (afterburner limit)
dT_AB_min = 50; % K minimum T06 - T04
%% ==== 3. Recover Goal 2 optimal hardware design ====
% Design vector: x = [Prc, Prf, beta, T04_nom, T06_nom, Cd_add]
x0 = [25, 1.5, 5.0, 1700, 2200, 0.02];
options = optimset('Display','off', ...
    'TolX',1e-4, 'TolFun',1e-6);
objFun = @(x) goal2_error_ST(x, targets, cond_to, cond_cr, params);
[x_opt, J_opt] = fminsearch(objFun, x0, options);
Prc_opt = x_opt(1);
Prf_opt = x_opt(2);
beta_opt = x_opt(3);
T04_nom = x_opt(4);
T06_nom = x_opt(5);
Cd_add_opt = x_opt(6);
params.Cd_add = Cd_add_opt; % store into params for cycle calls
fprintf('==== Goal 3: =====\n');
fprintf('Prc = %.3f\n', Prc_opt);
fprintf('Prf = %.3f\n', Prf_opt);
fprintf('beta = %.3f\n', beta_opt);
fprintf('Nominal T04 = %.1f K, T06 = %.1f K\n', T04_nom, T06_nom);
fprintf('Cd_add = %.4f\n', Cd_add_opt);
fprintf('Goal-2 objective J = %.4e\n\n', J_opt);
%% ==== 4. Define throttle search ranges (T04, T06) ====
% We sweep main-burner T04 and AB T06 over reasonable grids
T04_vec = linspace(T04_min, T04_max, 41); % 10 K steps
T06_vec = linspace(1800, T06_max, 81); % 10 K steps
%% ==== 5. Max thrust search at TAKEOFF (AB ON) ====
[res_to] = maxThrustForCondition(cond_to, true, ...
    Prc_opt, Prf_opt, beta_opt, T04_vec, T06_vec, dT_AB_min, params);
%% ==== 6. Max thrust search at CRUISE (AB ON) ====
[res_cr] = maxThrustForCondition(cond_cr, true, ...
    Prc_opt, Prf_opt, beta_opt, T04_vec, T06_vec, dT_AB_min, params);
%% ==== 7. Print summary ====
fprintf('\n==== MAX THRUST RESULTS (AB-MBTF, Goal-2 hardware) =====\n');
fprintf('\n--- Takeoff (Ground Roll) ---\n');
fprintf('Flight condition: M = %.2f, Ta = %.1f K, pa = %.1f kPa\n', ...
    cond_to.M, cond_to.Ta, cond_to.pa/1000);
fprintf('Max effective specific thrust = %.1f m/s (%.3f kN*s/kg)\n', ...
    res_to.STeff_max, res_to.STeff_max/1000);
fprintf('Core specific thrust (ST) = %.1f m/s (%.3f kN*s/kg)\n', ...
    res_to.ST_max, res_to.ST_max/1000);
fprintf('TSFC at max thrust = %.6e kg/(N*s)\n', res_to.TSFC_at_max);
fprintf('T04 (main burner) = %.1f K\n', res_to.T04_best);
fprintf('T06 (afterburner) = %.1f K\n', res_to.T06_best);

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fprintf('\n--- High-Altitude Cruise ---\n');
fprintf('Flight condition: M = %.2f, Ta = %.1f K, pa = %.1f kPa\n', ...
    cond_cr.M, cond_cr.Ta, cond_cr.pa/1000);
fprintf('Max effective specific thrust = %.1f m/s (%.3f kN·s/kg)\n', ...
    res_cr.STeff_max, res_cr.STeff_max/1000);
fprintf('Core specific thrust (ST) = %.1f m/s (%.3f kN·s/kg)\n', ...
    res_cr.ST_max, res_cr.ST_max/1000);
fprintf('TSFC at max thrust = %.6e kg/(N·s)\n', res_cr.TSFC_at_max);
fprintf('T04 (main burner) = %.1f K\n', res_cr.T04_best);
fprintf('T06 (afterburner) = %.1f K\n', res_cr.T06_best);
fprintf('\n=====');
%% ===== Local helper: sweep throttle to find max thrust =====
function res = maxThrustForCondition(cond, useAB, ...
    Prc, Prf, beta, T04_vec, T06_vec, dT_AB_min, params)
STeff_max = -Inf;
ST_max = NaN;
TSFC_best = NaN;
T04_best = NaN;
T06_best = NaN;
for i = 1:length(T04_vec)
    T04 = T04_vec(i);
    % For this T04, AB temperature must be at least T04 + dT_AB_min
    for j = 1:length(T06_vec)
        T06 = T06_vec(j);
        if T06 <= T04 + dT_AB_min
            continue; % violates minimum AB temperature rise
        end
        [TSFC, ST, ST_eff] = AB_MBTF_cycle(cond, useAB, ...
            Prc, Prf, beta, T04, T06, params);
        if ~isfinite(ST_eff) || ST_eff <= 0
            continue;
        end
        if ST_eff > STEff_max
            STEff_max = ST_eff;
            ST_max = ST;
            TSFC_best = TSFC;
            T04_best = T04;
            T06_best = T06;
        end
    end
end
end
res.STeff_max = STEff_max;
res.ST_max = ST_max;
res.TSFC_at_max = TSFC_best;
res.T04_best = T04_best;
res.T06_best = T06_best;
end

```